

**TED (21) – 3054
REVISION 2021**

QID : MODEL

**Reg. No.
Name :**

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE.
INTERNAL COMBUSTION ENGINES**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive Level
1	Write down the function of fuel feed pump.	M 1.01	R
2	Write down the function of fuel pressure regulator.	M 1.04	R
3	List two types of governors.	M2.02	R
4	Write down the function of fuel injector.	M 2.03	R
5	Write down the use of oil additives.	M 3.01	R
6	Write down the purpose of radiator.	M 3.04	R
7	Define the term Variable Time Valve Train.	M 4.01	R
8	Write down the material used for the construction of inlet manifold.	M 4.01	R
9	Write down the purpose of a turbocharger.	M 4.03	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive Level
1	Describe various air fuel ratios.	M 1.01	R
2	Explain the function of any three actuators in modern cars.	M 1.04	U
3	Describe types of petrol injection according to location of injector.	M 1.03	R
4	List the components of a conventional diesel fuel feed system.	M 2.01	R
5	Describe primary filter used in diesel engines.	M 2.02	R

6	Explain the specialty of splash lubrication system.	M 3.02	U
7	Explain the function of a thermostat.	M3.04	U
8	Explain “Electronic throttle body”.	M4.01	U
9	Explain the working of resonance type muffler.	M 4.02	U
10	Describe “waste gate turbo charger”.	M 4.04	R

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

		Module Outcome	Cognitive Level
1	Explain the function of any seven sensors used in modern petrol engines.	M 1.04	A
	OR		
2	Explain On-Board Diagnostics procedure of modern EFI Vehicles.	M 1.04	A
3	Explain the working of Jerk type fuel pump.	M 2.03	U
	OR		
4	Explain the working of a distributor type fuel pump.	M 2.03	U
5	Explain the functions of ECU in electronic diesel fuel system.	M 2.04	U
	OR		
6	Explain the working of electronic diesel fuel injector.	M 2.04	U
7	Explain dry sump lubrication system.	M 3.01	U
	OR		
8	Explain rotor type oil pump used in lubrication system.	M 3.02	U
9	Explain the working of pressure cap with a neat sketch	M 3.04	U
	OR		
10	Explain constructional details of water pump.	M 3.04	U
11	Explain Continuously Varying Valve Timing (CVVT).	M 4.01	U
	OR		
12	Explain the constructional details of a Turbocharger.	M4.04	U